

Honors Calculus III, Homework 1

Please upload solutions to **Problem 9 and Question 2** by **8pm on Friday, March 31**. You can use any results you want from lectures, provided you state them clearly, but make sure to write complete proofs. The remaining problems will not be graded, and you should not hand them in, but you are still expected to learn how to do them, unless it says “optional”.

Important: unless stated otherwise, to get full credit you need to show and explain your work. This applies both to homework and exams.

1. Spivak, *Calculus*, Chapter 24: Problems 1, 2, 3, 5, 9(*5pt), 16, 29.
2. If $n \in \mathbf{N}$ and $\alpha \in \mathbf{R}$ we define the binomial coefficient by the formula

$$\binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!},$$

which generalizes the usual definition when $\alpha \in \mathbf{N}$. For the rest of this problem we fix some $\alpha \in \mathbf{R}$.

- (a) (*5pt) Prove that

$$f(x) = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n$$

converges absolutely if $|x| < 1$.

- (b) (*5pt) Prove that $(1+x)f'(x) = \alpha f(x)$ whenever $|x| < 1$.
(c) (*5pt) Prove that $f(x) = (1+x)^\alpha$ whenever $|x| < 1$.